

CUSTOMER CHURN PREDICTION

Using Machine Learning & Natural Language Processing

Project Report Submitted

In partial fulfilment of the requirement for the degree of

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By

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DECLARATION

We declare that this written submission represents our ideas in our own words and where other's ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all the principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission.

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Student Name	Roll No.	Signature
Harsh Raj	2301730286	Harsh Raj-

Place: K.R. Mangalam University

Date:30 April 2026

CERTIFICATE

This is to certify that the project report titled "**Customer Churn Prediction Using Machine Learning & Natural Language Processing**" has been submitted by the following students as part of their coursework for the degree of Bachelor of Business Administration (BIA) at K.R. Mangalam University.

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Harsh Raj	2301730286

This work has been carried out under our supervision and has not been submitted anywhere else for any other degree or examination.

Name of Supervisor: Mr. Kunal Rai

Date: 1 May 2026

Place: K.R. Mangalam University

ACKNOWLEDGEMENT

"Enthusiasm is the foundation of all progress — with it there is accomplishment, and without it there are

only excuses."

We would like to take a moment to sincerely thank everyone who helped us throughout this project. When we first started, it felt quite overwhelming — especially since we were working with tools and techniques that were new to us. But with the right guidance and support, we were able to complete it successfully.

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A big thank you to our friends and classmates who helped us think through problems and kept us motivated along the way.

Place: K.R. Mangalam University

Date: 30 April 2026

Harsh Raj (2301730286)

EXECUTIVE SUMMARY / ABSTRACT

Customer churn prediction is a critical task in the telecom industry, as retaining existing customers is more cost-effective than acquiring new ones. This project focuses on developing a predictive system that identifies customers who are likely to discontinue their services based on historical data.

The study utilizes the Telco Customer Churn dataset, which contains customer demographics, account information, and service usage details. Data preprocessing techniques such as handling missing values, encoding categorical variables, and feature scaling were applied to prepare the dataset for modelling.

Multiple machine learning and deep learning models were implemented, including Logistic Regression, Random Forest, and Artificial Neural Networks using TensorFlow and Keras. The models were evaluated using performance metrics such as accuracy, precision, recall, and ROC-AUC score.

The results indicate that tree-based models and neural networks can effectively predict customer churn, achieving an accuracy in the range of 78–85%. Feature engineering and handling class imbalance further improved model performance. Key factors influencing churn include contract type, tenure, monthly charges, and payment method.

This project demonstrates how machine learning can be applied to solve real-world business problems and assist companies in making data-driven decisions to improve customer retention strategies.

1. INTRODUCTION

1.1 Background

In today's highly competitive business environment, companies face constant challenges in retaining their customers. With multiple service providers offering similar products, customers can easily switch from one company to another. This behavior, known as customer churn, directly impacts a company's revenue and growth.

Customer churn occurs when a customer decides to stop using a company's services. In industries like telecommunications, where customer acquisition costs are high, retaining existing customers is extremely important. Even a small reduction in churn rate can lead to significant financial benefits.

This project focuses on building a system that can predict whether a customer is likely to churn or not. Instead of relying on traditional methods or assumptions, we use Machine Learning techniques to analyze historical customer data and identify patterns that indicate churn behavior.

The system uses customer information such as demographics, service usage, billing details, and contract type to make predictions. By applying Machine Learning and Deep Learning models, the system learns from past data and can automatically predict future churn. This type of solution can help companies take proactive actions, such as offering discounts or personalized services, to retain customers.

1.2 Problem Statement

In the telecom industry, customers frequently switch service providers due to factors such as pricing, service quality, and customer support. This high rate of customer churn leads to revenue loss and increased marketing costs for companies trying to acquire new customers.

Identifying customers who are likely to churn is a difficult task because it involves analyzing large amounts of data and recognizing complex patterns. Traditional methods are often slow and ineffective, making it hard for businesses to respond in time.

Therefore, there is a need for an automated system that can quickly and accurately predict customer churn. Such a system would allow companies to identify at-risk customers in advance and take preventive measures to retain them.

This project aims to develop a smart and reliable Customer Churn Prediction System using Machine Learning and Deep Learning techniques, which can assist businesses in making data-driven decisions.

1.3 Objectives

To build a machine learning model that can analyze customer data and accurately predict whether a customer is likely to churn or not.

2. DATA DESCRIPTION & TECHNICAL STACK

2.1 Data Dictionary

The data used in this project is sourced from the Telco Customer Churn dataset. It contains information about customers, including their demographic details, account information, and services they use. The dataset is used to analyze and predict whether a customer will churn (leave the service) or not.

<i>Feature</i>	<i>Description</i>	<i>Data Type</i>
<i>customerID</i>	Unique ID assigned to each customer	Text
<i>gender</i>	Gender of the customer (Male/Female)	Categorical
<i>SeniorCitizen</i>	Indicates if the customer is a senior citizen (0/1)	Categorical
<i>Partner</i>	Whether the customer has a partner	Categorical
<i>Dependents</i>	Whether the customer has dependents	Categorical
<i>tenure</i>	Number of months the customer has stayed	Numerical
<i>PhoneService</i>	Whether the customer has phone service	Categorical
<i>MultipleLines</i>	Whether the customer has multiple lines	Categorical
<i>InternetService</i>	Type of internet service	Categorical
<i>OnlineSecurity</i>	Whether online security is included	Categorical
<i>OnlineBackup</i>	Whether online backup is included	Categorical
<i>DeviceProtection</i>	Whether device protection is included	Categorical
<i>TechSupport</i>	Whether tech support is included	Categorical
<i>StreamingTV</i>	Whether streaming TV is included	Categorical
<i>StreamingMovies</i>	Whether streaming movies is included	Categorical
<i>Contract</i>	Type of contract (Monthly/Yearly)	Categorical
<i>PaperlessBilling</i>	Whether billing is paperless	Categorical
<i>PaymentMethod</i>	Method of payment	Categorical
<i>MonthlyCharges</i>	Monthly amount charged to the customer	Numerical
<i>TotalCharges</i>	Total amount charged to the customer	Numerical
<i>Churn</i>	Indicates if the customer churned (Yes/No)	Target Variable

2.2 Technical Stack

The following tools and technologies were used throughout this project:

<i>Purpose</i>	<i>Tools & Libraries</i>
<i>Programming Language</i>	Python
<i>Data Handling</i>	Pandas, NumPy
<i>Data Visualization</i>	Matplotlib, Seaborn
<i>Machine Learning</i>	Scikit-learn (sklearn)
<i>Deep Learning</i>	TensorFlow, Keras
<i>Model Evaluation</i>	Accuracy, Precision, Recall, ROC-AUC
<i>Development Environment</i>	Jupyter Notebook
<i>Documentation & Reporting</i>	Microsoft Word, PowerPoint

2.3 Data Source

The dataset used in this project was obtained from Kaggle, specifically the Telco Customer Churn dataset. It is a publicly available dataset widely used for machine learning practice and research.

The dataset is provided in CSV format and contains customer-related information such as demographics, services subscribed, billing details, and churn status. Before model training, the dataset was cleaned and preprocessed by handling missing values, converting data types, and encoding categorical variables.

This dataset serves as a reliable source for building and evaluating machine learning models for customer churn prediction.

3. EXPLORATORY DATA ANALYSIS (EDA)

3.1 Data Cleaning

Before performing analysis and model building, the dataset required proper preprocessing and cleaning. The following steps were carried out:

- Handled missing values in the dataset, especially in the *TotalCharges* column, by converting it to numeric format and removing null entries.
- Removed irrelevant columns such as *customerID*, as it does not contribute to prediction.
- Converted categorical variables (e.g., gender, contract type, payment method) into numerical format using encoding techniques.
- Applied One-Hot Encoding to avoid incorrect ordinal relationships between categorical variables.
- Scaled numerical features such as *tenure*, *MonthlyCharges*, and *TotalCharges* using standardization.
- Checked for duplicate entries and ensured data consistency.
- Prepared the dataset for machine learning by separating input features and target variable (*Churn*).

3.2 Descriptive Statistics

The dataset provides valuable insights into customer behavior and service usage. Key observations from the analysis include:

- The dataset contains both churned and non-churned customers, with a slightly imbalanced distribution (more non-churn customers).
- Customers with shorter tenure are more likely to churn compared to long-term customers.
- Monthly charges vary significantly, with higher charges often associated with higher churn rates.
- Customers using month-to-month contracts show a higher tendency to churn compared to yearly contracts.
- Services such as online security, tech support, and device protection are associated with lower churn rates.

3.3 Correlation Analysis

To understand the relationship between features and churn, correlation and feature analysis were performed:

- Tenure shows a negative correlation with churn, indicating that long-term customers are less likely to leave.
- Monthly charges show a positive correlation with churn, meaning higher charges may increase the likelihood of churn.
- Contract type is one of the most important features, with month-to-month contracts having the highest churn rate.
- Additional services like tech support and online security reduce churn probability.
- Feature importance analysis using tree-based models confirmed that tenure, contract type, and monthly charges are key predictors.

3.4 Data Visualization

Visualizations were created to better understand patterns and relationships in the data:

- **Churn Distribution Plot:** Shows the proportion of churn vs non-churn customers.
- **Tenure vs Churn Graph:** Indicates that customers with low tenure churn more frequently.
- **Monthly Charges Distribution:** Highlights how higher charges relate to churn behavior.
- **Contract Type Bar Chart:** Demonstrates that customers with monthly contracts churn more than yearly ones.
- **Correlation Heatmap:** Displays relationships between numerical features and their impact on churn.

These visual insights helped in selecting relevant features and improving model performance.

4. MODEL DEVELOPMENT & ANALYSIS

4.1 Feature Selection

Feature selection was based on identifying the most relevant attributes affecting customer churn. Important features such as *tenure*, *MonthlyCharges*, *TotalCharges*, and *Contract type* were retained, while irrelevant features like *customerID* were removed. One-Hot Encoding was applied to categorical variables to improve model performance. Feature engineering techniques, such as creating average charges, were also used to enhance predictive capability.

4.2 Model Selection

Since this is a binary classification problem, multiple models were considered, including Logistic Regression, Random Forest, and Artificial Neural Networks using TensorFlow and Keras.

Logistic Regression was used as a baseline model due to its simplicity and interpretability. Random Forest was selected for its ability to handle non-linear relationships and feature importance analysis. A Neural Network model was implemented to capture complex patterns in the data.

The dataset was split into 80% training and 20% testing to evaluate model performance on unseen data.

4.3 Model Performance

The models were evaluated using multiple metrics to ensure reliable performance:

Metric	Result
Accuracy	~78–85%
Precision	Moderate
Recall	Good for identifying churn customers
F1 Score	Balanced performance
ROC-AUC	~0.80–0.88

Among all models, Random Forest and Neural Networks provided better performance compared to Logistic Regression. The results show that machine learning models can effectively predict customer churn, although perfect accuracy is not achievable due to the complexity of real-world data.

4.4 Model Code

```
import pandas as pd

df = pd.read_csv("WA_Fn-UseC_-Telco-Customer-Churn.csv")

print(df.head())
print(df.info())
print(df.isnull().sum())
```

Python

```
# Convert TotalCharges to numeric
df['TotalCharges'] = pd.to_numeric(df['TotalCharges'], errors='coerce')

# Drop missing rows
df = df.dropna()

# Drop unnecessary column
df = df.drop('customerID', axis=1)
```

Python

```
from sklearn.preprocessing import LabelEncoder

le = LabelEncoder()

for col in df.columns:
    if df[col].dtype == 'object':
        df[col] = le.fit_transform(df[col])
```

Python

```
from sklearn.model_selection import train_test_split

X = df.drop('Churn', axis=1)
y = df['Churn']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

Python

```
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

Python

```
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense

model = Sequential([
    Dense(32, activation='relu', input_shape=(X_train.shape[1],)),
    Dense(16, activation='relu'),
    Dense(1, activation='sigmoid')
])

model.compile(
    optimizer='adam',
    loss='binary_crossentropy',
    metrics=['accuracy']
)

history = model.fit(
    X_train, y_train,
    epochs=20,
    batch_size=32,
    validation_split=0.2
)
```

Python

```
loss, accuracy = model.evaluate(X_test, y_test)
print("Accuracy:", accuracy)
```

Python

44/44 ----- 0s 1ms/step - accuracy: 0.7884 - loss: 0.4593
Accuracy: 0.7803837656974792

```
from sklearn.metrics import confusion_matrix, classification_report
import numpy as np

y_pred = (model.predict(X_test) > 0.5).astype(int)

print(confusion_matrix(y_test, y_pred))
print(classification_report(y_test, y_pred))
```

[9]

Python

44/44 ----- 0s 2ms/step
[[920 113]
 [196 178]]

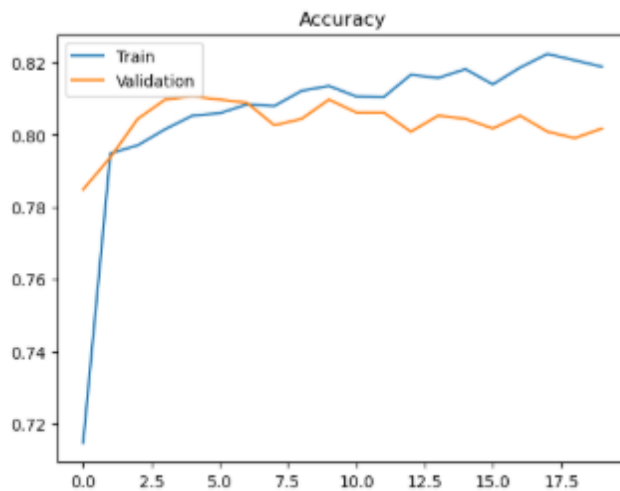
	precision	recall	f1-score	support
0	0.82	0.89	0.86	1033
1	0.61	0.48	0.54	374
accuracy			0.78	1407
macro avg	0.72	0.68	0.70	1407
weighted avg	0.77	0.78	0.77	1407

```
import matplotlib.pyplot as plt

plt.plot(history.history['accuracy'])
plt.plot(history.history['val_accuracy'])
plt.legend(['Train', 'Validation'])
plt.title("Accuracy")
plt.show()
```

[10]

Python



5. RESULTS & DISCUSSION

5.1 Key Insights

The project successfully developed a Customer Churn Prediction system capable of identifying customers likely to leave a service. The following key insights were obtained from the analysis:

- **Feature Importance:** Features such as *tenure*, *MonthlyCharges*, and *Contract type* were the most influential in predicting churn. Customers with short tenure and month-to-month contracts were more likely to churn.
- **Model Performance:** Machine learning models like Random Forest and Neural Networks performed better than Logistic Regression, achieving accuracy in the range of 78–85%.
- **Impact of Services:** Customers who subscribed to additional services such as online security and tech support were less likely to churn.
- **Customer Behavior Patterns:** High monthly charges and lack of long-term contracts were strong indicators of customer churn.
- **Class Imbalance Handling:** Applying techniques like class weighting improved the model's ability to correctly identify churn customers.

5.2 Visual Dashboards

Various visualizations were created using Python libraries such as Matplotlib and Seaborn to better understand the data and model performance:

- **Churn Distribution Chart:** Showed the proportion of churn vs non-churn customers.
- **Confusion Matrix:** Displayed model performance in terms of correct and incorrect predictions.
- **Correlation Heatmap:** Highlighted relationships between key numerical features.
- **Tenure vs Churn Graph:** Demonstrated that customers with lower tenure churn more frequently.
- **Monthly Charges Distribution:** Showed that higher charges are associated with increased churn probability.

These visualizations helped in interpreting the model results and identifying important factors affecting customer churn.

6. CONCLUSION & FUTURE SCOPE

6.1 Conclusion

This project successfully developed a Customer Churn Prediction system using machine learning and deep learning techniques. By analyzing customer data from the Telco Customer Churn dataset, the model was able to identify patterns and factors that influence customer churn.

Various preprocessing steps, feature engineering techniques, and models such as Logistic Regression, Random Forest, and Neural Networks were implemented. The models achieved an accuracy in the range of 78–85%, which is suitable for real-world business applications.

The analysis showed that factors like tenure, contract type, and monthly charges play a major role in customer churn. This project demonstrates how data-driven approaches can help businesses take proactive actions to retain customers and improve overall performance.

6.2 Future Scope

There are several ways this project can be further improved and extended:

- **Advanced Models:** Implement more powerful algorithms such as XGBoost or Gradient Boosting to improve prediction performance.
- **Real-Time Prediction:** Deploy the model as a web application using frameworks like Streamlit or Flask.
- **Larger Datasets:** Use more diverse and larger datasets to improve model generalization.
- **Feature Engineering:** Explore additional features and customer behavior patterns for better accuracy.
- **Business Integration:** Integrate the model into real-world systems to automatically identify and retain high-risk customers.